

Pleural Effusions — Bedside Reference

Tap any effusion >1 cm (US/lateral decubitus). Two questions: transudate vs exudate (Light's), and — if parapneumonic — does it need drainage?
Adapted from TeachIM.org (Deshpande/Murphy; Zhang, Apr 2022), CC BY-NC 4.0.

Light's criteria — exudate if ANY one

Criterion	Exudate if
Protein ratio	pleural/serum protein >0.5
LDH ratio	pleural/serum LDH >0.6
LDH absolute	pleural LDH >2/3 serum ULN

Sens ~98%, spec ~83%. Diuretics misclassify ~25% of transudates
→ if it looks like a transudate, check **serum–pleural albumin gradient >1.2** = transudate. High pleural RBCs falsely raise LDH.

Differential

Transudate	Exudate
HF, renal/nephrotic, cirrhosis (hepatic hydrothorax), hypoalbuminemia, PD, atelectasis, trapped lung	Parapneumonic/infection, malignancy, PE, autoimmune (RA/SLE/GPA/Sjögren/sarcoid), chylothorax, hemothorax, GI (esoph perf, pancreatitis)

Targeted fluid studies

Finding	Points to
PMN predominant	Bacterial parapneumonic / empyema
Lymphocyte predom.	Malignancy, TB, autoimmune, chylo, post-CABG, chronic
ADA >40	TB pleurisy (>90% sens/spec)
TG >110	Chylothorax
Pleural:blood Hct >0.5	Hemothorax
Amylase ↑	Esoph rupture, pancreatitis
Cytology / flow	Malignancy (flow if leuk/lymphoma)

Parapneumonic management

Category	Features	Mgmt
Uncomplicated	no loc, pH>7.2, nl glucose	Antibiotics alone
Complicated	loculations OR pH<7.2 OR low glucose	Abx + DRAIN
Empyema	pus OR +Gram/culture	Abx + drain; VATS/decort if needed

Drain: small-bore ≤14F first-line (BTS 2023); 24–32F for heavy loculation/pus. **tPA + DNase** (MIST-2) improves drainage, reduces surgical referral — **no mortality benefit**.

Malignant effusion

- Cytology sens ~65% (1st tap) → ~92% (2nd) → ~97% (3rd); serial large-volume taps raise yield
- Send cytology on initial work-up of any uncertain effusion (BTS)
- Recurrent symptomatic MPE: indwelling pleural catheter OR talc pleurodesis (both 1st-line, ATS/STS/STR 2018); IPC+talc ↑ pleurodesis

Pearls

- Light's third criterion uses the serum LDH **ULN**, not the patient's serum LDH
- Pleural pH must be drawn anaerobically in a blood-gas syringe (air/lidocaine skew it)
- Low pleural glucose (<60): infection, malignancy, RA/SLE, esophageal rupture, TB
- Bloody effusion w/o trauma: malignancy, PE, or asbestos